MULTI-PERIOD HOUSE ALLOCATION AS A PUBLIC GOOD PROBLEM

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Abstract. We prove existence of envy-free and Pareto efficient allocation in a multi-period indivisible goods assignment problem. Since Birkoff-von-Neumann fails in this environment, the allocation problem cannot be reduced to individual-level consumption problem where total demand cannot exceed supply. Therefore, we have to treat economy-wide allocations as a public good. We provide a market-based implementation by showing that such an allocation can be supported as a Lindahl equilibrium. We then look at a large economy and show that for large enough economies, the incentives to misrepresent agents' preferences vanish.